



Monday 12 December 2022
09:00-10:30 GMT
(Duration: 1 hours 30 minutes)
Additional time: 30 minutes
Timed exam – fixed start time

DEGREE OF MSc

Research Professional Skills (M)
COMPSCI5092

(Answer All Questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. For each of the following multiple-choice questions, select the correct answer.

Note: in the following multiple-choice questions that +1 mark will be awarded for each correct answer, -0.5 for each incorrect answer and 0 marks for any answer left blank:

(a) Information sources can be categorised in different ways. Which of these statements is true?

- a. The difference between review articles and research articles is that review articles do not follow any defined methodology
- b. A paper that presents the detailed measurements of the screen size of different mobile phones may be primary or secondary research
- c. It is often not worth spending the additional time needed to conduct primary research

(b) The law of confidential information...

- a. Applies only to people employed by a commercial organisation
- b. Primarily concerns itself with information that someone has been told in confidence
- c. Prevents someone revealing information they have received in confidence

(c) An annotated bibliography...

- a. Must include all the details of the paper
- b. Only contains peer reviewed research articles
- c. Is often included at the end of an article

(d) In considering the nature of primary and secondary data...

- a. Secondary data may introduce some sort of bias or distortion
- b. Newspaper articles are secondary sources that never report primary data
- c. Primary data provides better evidence for supporting your conclusion than secondary data

[4]

(e) Describe four requirements for a database to be considered protected by copyright / intellectual property rights.

[4]

(f) Discuss three reasons why societal issues in IT often require existing legislation be amended or new legislation be introduced.

[3]

2. A researcher is investigating whether first year Computing Science students who also study Physics show greater improvement in their programming abilities by the end of the year than those who do not.

The researcher has planned a simple experiment to test this belief. First year Computing Science students will complete a multiple-choice programming quiz at the end of the academic year. The researcher will then compare the mean score from each group of students (i.e., Physics vs non-Physics) to see if the mean is higher for Physics students.

Critique this experiment design by explaining three ways in which the proposed experiment may be unable to answer the research question. Propose changes to address any issues you identified.

[9]

3. You are planning an experiment to investigate if paired programming helps novice Java programmers solve lab exercises during a programming course.

Note: Pair programming is a software development technique in which two programmers work together at a single workstation. One programmer, the driver, writes code while the other, the observer, reviews each line of code as it is typed in.

(a) Discuss the advantages of both within-subject and between-subject designs for this experiment. Clearly link your identified advantages to the given experimental scenario.

Indicate which approach you would choose and justify your choice.

[6]

(b) Identify one quantitative and one qualitative dependent variable for this experiment. Justify your choice of each chosen variable.

[4]

(c) Describe one closed and one open-ended question that you could use to capture your qualitative dependent variable from part (b).

[2]

(d) Your experiment will require ethical approval because it involves human participants. Identify two potential ethical concerns with this research

[3]

4. You have developed a warehouse assistance app for an augmented reality headset, to be used by workers in a parcel delivery company's warehouse. The tool you have developed provides workers with information to improve their logistical operations including: directions/route suggestions for navigating the warehouse, packaging instructions for parcels, time management information, and additional information about some parcels (e.g., reminders to handle a parcel with care, warnings if a parcel is too heavy to be safely lifted by one person, etc).

You believe that using this tool will help increase worker's productivity compared to workers without the warehouse assistance tool.

Propose an experiment to investigate this belief. You should cover all aspects of a research project, except performing a literature survey and writing a research proposal.

[25]